

Exercise 1.

Two fair six-sided dice are rolled. Let the random variable X denote the product of the 2 dice. Find the probability of: (a) $P(X \leq 2)$, (b) $P(X \leq 35)$.

Exercise 2.

Three fair coins are tossed. Let the random variable X denote the number of heads. Write out the pmf and cdf of X .

Exercise 3.

Suppose that the distribution function of X is given by

$$F(b) = \begin{cases} 0, & b < 0, \\ \frac{b}{4}, & 0 \leq b < 1, \\ \frac{1}{2} + \frac{b-1}{4}, & 1 \leq b < 2, \\ \frac{11}{12}, & 2 \leq b < 3, \\ 1, & 3 \leq b. \end{cases}$$

Find $P(X = i), i = 1, 2, 3$.

Exercise 4.

Let X denote a random variable that takes on any of the values -1 , 0 , and 2 . The pmf of X is given by

$$p(x) = \begin{cases} 1/5, & \text{if } x = -1, \\ 1/5, & \text{if } x = 0, \\ 3/5, & \text{if } x = 2, \\ 0, & \text{otherwise.} \end{cases}$$

Compute:

- ❶ $E[X^3]$,
- ❷ $E[2X^2]$,
- ❸ $E[4X^2 - 1]$.

Exercise 5.

If $E[X] = 1$ and $Var(X) = 5$, find

- Ⓐ $E[(2 + X)^2]$
- Ⓑ $Var(4 + 3X)$

Exercise 6.

Let X be the number of 6 obtained when roll four fair 6-sided dice simultaneously. Its pmf is.

x	0	1	2	3	4
$p(x)$	0.4823	0.3858	0.1157	0.0154	0.0008

Find its cdf $F(x)$.

Exercise 7.

The probability mass function of a discrete random variable X is given by

$$p(x) = P(X = x) = \begin{cases} kx, & \text{if } x = 1, 2, \dots, n \\ 0, & \text{otherwise.} \end{cases}$$

- Ⓐ $k = ?$
- Ⓑ $E(X) = ?$

Exercise 8.

The discrete random variable X has probability mass function given by

$$p(x) = \begin{cases} \frac{2}{10}, & \text{if } x = -1, \\ \frac{1}{10}, & \text{if } x = 0, \\ a, & \text{if } x = 1, \\ \frac{4}{10}, & \text{if } x = 2, \\ b, & \text{if } x = 3, \\ 0, & \text{otherwise.} \end{cases}$$

Given that $E(X) = 1.5$ then (a) $a = ?$ and $b = ?$ (b) $E(X^2) = ?$